MASTER 2 · SESSION 1 / 7

Causal Inference

Session 1 — Causal questions and fair comparisons

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Albert School · Master 2

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ALBERT SCHOOL

How this session is built

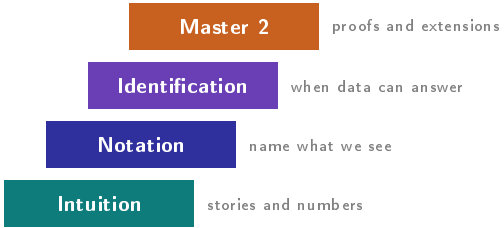
Introduction

OVERVIEW

Every formula today translates a sentence you will already have agreed with.

We climb four steps. We start with a school, some students and a few numbers. Only when the idea is clear do we give it a symbol, then ask when data can answer it, and finally prove things at Master 2 level.

Look at the badge in the top-right corner of each slide: it tells you which step you are on.



Remember. Intuition first, symbols second, proofs last. Never the other way round.

By the end of today you should be able to

Introduction OVERVIEW

1. Tell a **causal** question apart from a prediction or a description.
2. Explain why a raw gap between two groups is usually **not** a causal effect.
3. Write potential outcomes, ATE, ATT and CATE, and compute them on a small table.
4. Draw a causal graph and spot a confounder, a mediator and a collider.
5. Build a fair comparison by standardisation and state the three assumptions it needs.
6. Prove identification results, derive omitted-variable bias, compute sharp bounds.

Remember. Items 1–2 are intuition, 3–4 notation, 5 identification, 6 Master 2 mathematics.

Session 1 roadmap

1. Why ask a causal question?	**Intuition**
2. The missing counterfactual	Intuition
3. From story to notation	Notation
4. Drawing causal stories	Notation
5. Building a fair comparison	Identification
6. The Master 2 toolkit	Master 2
7. When assumptions fail	Master 2
8. Practice and wrap-up	Synthesis

Does tutoring improve exam results?

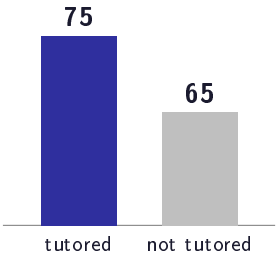
Why ask a causal question?

INTUITION

A school runs an optional tutoring programme before the final exam.

Students who attended score **75** on average. Students who did not score **65**. The principal wants to double the programme's budget.

Before spending the money, she really wants to know: *would these students have scored less without tutoring?*



(axis starts at 50)

Remember. A gap between two groups is a fact. Saying tutoring *caused* it is a claim.

Three questions that sound alike

Why ask a causal question? INTUITION

The same spreadsheet can be asked three very different questions.

Question	Type
Which students are likely to fail the exam?	**Prediction**
Do tutored students score higher than the others?	**Description**
Would a student score higher *if we gave them* tutoring?	**Causal**

Only the third question tells the principal what will happen when she *acts*.

Remember. A decision to change something needs a causal question.

Predicting is not the same as intervening

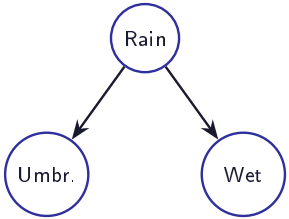
Why ask a causal question?

INTUITION

Umbrellas predict rain very well.

Seeing many umbrellas in the street tells you it is probably raining. But banning umbrellas will not make the sun come out.

In the same way, the number of library visits may predict exam scores well. Forcing every student into the library is a different question: it asks what happens when we *change* visits.



umbrellas and wet streets
are correlated, not causally linked

Remember. A good predictor need not be a good lever.

Causal questions are everywhere

Why ask a causal question? INTUITION

The tutoring example is a template for the questions you will meet in practice.

Marketing
Did the 20% discount increase sales, or were customers who received it already loyal buyers?

Finance
Does joining a stock index raise a firm's price, or do already-rising firms get added?

Public policy
Do job-training programmes raise wages, or do the most employable people sign up?

Remember. Each case has the same structure: who receives the treatment is not random.

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One student, two possible worlds

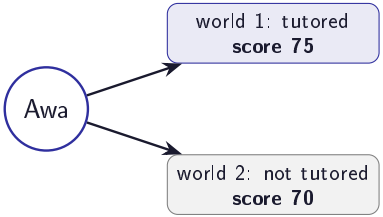
The missing counterfactual

INTUITION

Think about Awa, a single student.

In one world she attends tutoring and scores 75. In another world, identical in every other way, she skips it and scores 70.

The effect of tutoring *for Awa* is the difference between the two worlds: $75 - 70 = 5$ points.



Remember. A causal effect compares two worlds for the *same* person.

The catch: we only ever see one world

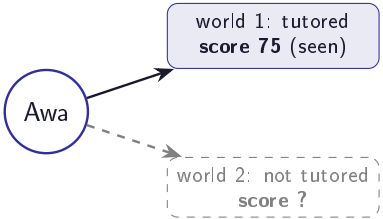
The missing counterfactual

INTUITION

Awa did attend tutoring. We see 75.

The score of 70 lives in a world that never happened. No survey, no bigger dataset and no better algorithm will ever record it.

Statisticians call this the **fundamental problem of causal inference** (Holland, 1986).



Remember. Causal inference is a missing-data problem: half of the answer is never observed.

The class as the school sees it

The missing counterfactual INTUITION

Six students. For each one, only the score in the world that happened is known.

Student	Tutored?	Score if tutored	Score if not tutored
Awa	yes	75	?
Léo	yes	76	?
Inès	yes	74	?
Moussa	no	?	62
Chloé	no	?	66
Karim	no	?	67

Tutored average: 75. Untutored average: 65. The gap is 10 points.

Remember. Every real dataset looks like this: one column is always half empty.

The class as an oracle would see it

The missing counterfactual

INTUITION

Suppose, just for learning, that we could see both worlds.

Student	Tutored?	Score if tutored	Score if not tutored	Effect
Awa	yes	75	70	5
Léo	yes	76	72	4
Inès	yes	74	68	6
Moussa	no	71	62	9
Chloé	no	72	66	6
Karim	no	70	67	3

Violet numbers are invented for teaching: in reality they are never seen. Effects range from 3 to 9 points.

Remember. Effects differ across students; no single number describes everyone.

The naive comparison overstates the effect

The missing counterfactual

INTUITION

The raw gap says 10 points. The oracle table says tutored students gained 5 on average.

Average effect for the three tutored students: $(5 + 4 + 6)/3 = 5$.

So where do the other 5 points of the 10-point gap come from?

Look at the violet column: *without* tutoring, Awa, Léo and Inès would have averaged 70. The other three averaged 65. The tutored students were already ahead.

Remember. Part of the raw gap was there before any tutoring took place.

Why the volunteers were already ahead

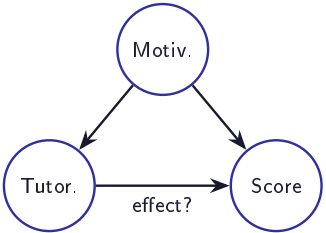
The missing counterfactual

INTUITION

Tutoring was optional. Who signs up for optional extra work?

Motivated students. And motivated students also revise more on their own, so they would score higher anyway.

Motivation pushes *both* towards tutoring *and* towards a higher score. A variable that does both is called a **confounder**.



Remember. A shared cause can produce a gap even if tutoring did nothing.

The 10-point gap, taken apart

The missing counterfactual

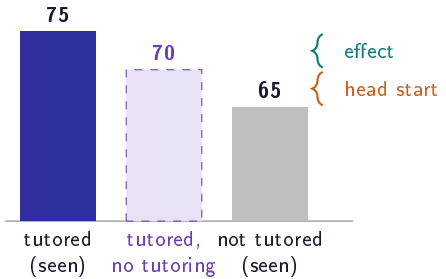
INTUITION

Two different things add up to the gap.

From 65 to 70: the **head start** of the volunteers, which exists without any tutoring. We will call it *selection bias*.

From 70 to 75: what tutoring really **added** for those who took it.

$10 = 5$ (head start) $+ 5$ (effect).



Remember. Observed gap = effect of treatment + differences that existed anyway.

Checkpoint

The missing counterfactual INTUITION

Discuss with your neighbour for three minutes.

1. A gym finds that members who hire a personal trainer lose more weight. Name one reason, other than the trainer, that could explain the gap.
2. In which direction does that reason bias the naive comparison?
3. What single piece of information would you need to measure the true effect for one member? Why can you never get it?

Remember. The whole course is about finding credible stand-ins for the missing world.

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Giving names to what we already see

From story to notation NOTATION

Nothing new here: we only attach letters to the ingredients of the story.

Symbol	Meaning	In the school
i	a unit	one student (Awa)
$D_i \in \{0, 1\}$	treatment received	1 if tutored, 0 otherwise
Y_i	observed outcome	exam score, 0–100
X_i	characteristics measured before treatment	baseline preparation
population	the people we want to learn about	students eligible this term

Remember. A precise question starts with precise definitions of unit, treatment, outcome and population.

Potential outcomes are the two columns of the oracle table

From story to notation NOTATION

"Score if tutored" and "score if not tutored" get their own symbols.

$Y_i(1)$ = score of student i if tutored, $Y_i(0)$ = score of student i if not tutored.

For Awa: $Y_{\text{Awa}}(1) = 75$ and $Y_{\text{Awa}}(0) = 70$. Her individual effect is

$$\tau_i = Y_i(1) - Y_i(0), \qquad \tau_{\text{Awa}} = 5.$$

Both potential outcomes exist *before* treatment is decided; treatment only decides which one we get to see.

Remember. $Y_i(1)$ and $Y_i(0)$ are the two worlds; τ_i is the difference between them.

The observed score is a switch

From story to notation NOTATION

Treatment decides which of the two columns reaches the spreadsheet.

$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0)$$

Check with Awa: $D = 1$, so $Y = 1 \cdot 75 + 0 \cdot 70 = 75$.
Check with Moussa: $D = 0$, so $Y = 0 \cdot 71 + 1 \cdot 62 = 62$.

This link between the story and the data is called **consistency**. It will be one of our assumptions.

Remember. The formula selects the outcome that occurred; the other one is lost.

The average treatment effect

From story to notation NOTATION

Individual effects are unobservable, so we aim for their average.

$$\text{ATE} = E\big[Y(1) - Y(0)\big]$$

$E[\cdot]$ simply means "average over the population". In the oracle table:

$$\text{ATE} = \frac{5 + 4 + 6 + 9 + 6 + 3}{6} = 5.5.$$

An ATE of 5.5 does *not* mean every student gains 5.5 points: Karim gains 3, Moussa gains 9.

Remember. ATE answers: what if we tutored everyone, compared with no one?

Whose effect? ATT and ATU

From story to notation NOTATION

The principal may ask about current participants, or about those who stayed away.

$$\text{ATT} = E\big[Y(1) - Y(0) \mid D = 1\big], \qquad \text{ATU} = E\big[Y(1) - Y(0) \mid D = 0\big].$$

The vertical bar reads "among students with". In the oracle table:

$$\text{ATT} = \tfrac{5+4+6}{3} = 5, \qquad \text{ATU} = \tfrac{9+6+3}{3} = 6, \qquad \text{ATE} = \tfrac{1}{2}\cdot 5 + \tfrac{1}{2}\cdot 6 = 5.5.$$

The students who stayed away would actually have gained more.

Remember. State whose effect you want before choosing a method.

Effects for groups: CATE

From story to notation

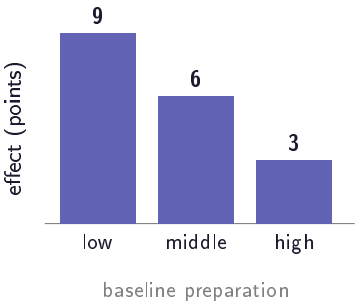
NOTATION

Tutoring may help weakly prepared students more than students already near 100.

The average effect among students with characteristics $X = x$ is

$$\tau(x) = E\big[Y(1) - Y(0) \mid X = x\big].$$

This is the **conditional** average treatment effect. It is still an average, not an individual effect.



Remember. Targeting support starts with conditional averages.

The naive gap, now in symbols

From story to notation

NOTATION

Earlier we split 10 points into 5 + 5. Here is the same split for any population.

Add and subtract the tutored group's untreated average $E[Y(0) \mid D = 1]$:

$$\underbrace{E[Y \mid D = 1] - E[Y \mid D = 0]}_{\text{raw gap: } 75-65=10} = \underbrace{E[Y(1) - Y(0) \mid D = 1]}_{\text{ATT: } 75-70=5} + \underbrace{E[Y(0) \mid D = 1] - E[Y(0) \mid D = 0]}_{\text{selection bias: } 70-65=5}.$$

The first term is what we want. The second compares the groups in the world *without* tutoring.

Remember. The raw gap equals the ATT only when the selection-bias term is zero.

When does selection bias vanish?

From story to notation NOTATION

The bias term asks: would the two groups have scored the same without tutoring?

If who gets tutoring has nothing to do with how well students would do anyway, then

$$E[Y(0) \mid D = 1] = E[Y(0) \mid D = 0] \quad \Longrightarrow \quad \text{raw gap} = \text{ATT}.$$

With volunteers this fails, because motivation drives both. If a **lottery** decides who gets tutoring, treatment cannot depend on $Y(0)$: that is next session's topic.

Today we ask a different question: without a lottery, can we still build a fair comparison?

Remember. Every method in this course is a way of making the selection-bias term disappear.

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An arrow is a claim about what changes what

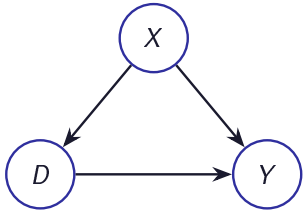
Drawing causal stories

NOTATION

Draw $A \to B$ only if changing A could change B directly.

A causal graph is a drawing of our story. It contains no numbers, only assumptions that everyone in the room can inspect and challenge.

Here X is baseline preparation, D tutoring and Y the exam score.



Remember. A missing arrow is also an assumption, often the strongest one.

Confounder: a common cause

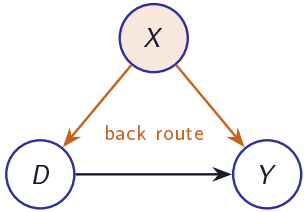
Drawing causal stories

NOTATION

Preparation X affects both who gets tutoring and the score.

There are two routes from D to Y in the graph: the causal arrow $D \to Y$, and the back route $D \leftarrow X \to Y$. The back route carries association but no effect.

Comparing students with the *same* X closes the back route.



Remember. Adjust for confounders.

Mediator: the channel of the effect

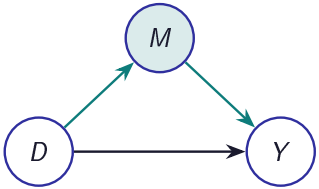
Drawing causal stories

NOTATION

Tutoring may work by improving study habits M, which then raise scores.

M is created by the treatment and carries part of its effect. If we compare students with the same study habits, we switch off that channel and hide part of what tutoring does.

That comparison answers a narrower question (a direct effect) that needs its own assumptions.



Remember. Do not adjust for a variable caused by treatment when you want the total effect.

Collider: a common consequence

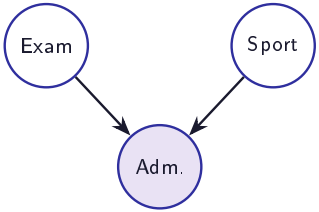
Drawing causal stories

NOTATION

A selective programme admits students with strong exam ability *or* strong sports ability.

In the general population the two abilities are unrelated. Among admitted students, someone weak at exams must be strong at sport to get in.

Looking only at admitted students therefore creates a negative relationship that does not exist in the population.



Remember. Conditioning on a common effect creates an association.

Seeing the collider effect

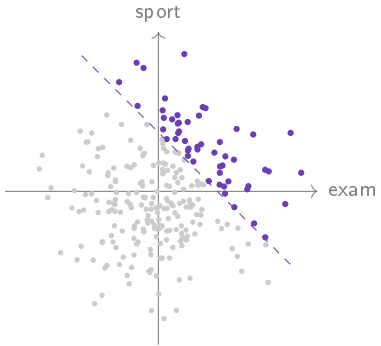
Drawing causal stories

NOTATION

Each dot is a simulated applicant. The two abilities are drawn independently.

Grey cloud: no relationship. Violet dots: the admitted students, whose total ability is high. Within the violet group, higher exam ability goes with lower sports ability.

Nothing causal happened: only selection.



Remember. A selected sample can show relationships its population does not have.

Three roles, three decisions

Drawing causal stories NOTATION

The same question, "should I control for this variable?", has three different answers.

Role	Shape	Example	Adjust for total effect?
Confounder	$D \leftarrow X \rightarrow Y$	baseline preparation	**yes**
Mediator	$D \rightarrow M \rightarrow Y$	study habits after tutoring	**no**
Collider	$D \rightarrow C \leftarrow Y$	being selected into a sample	**no**

"Control for everything" is therefore not a safe rule: it can remove a real effect or create a fake one.

Remember. The graph, not the regression software, decides which controls belong.

Seeing versus doing

Drawing causal stories

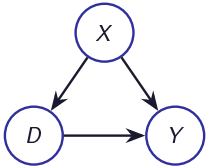
NOTATION

Observing who chose tutoring is not the same as assigning it.

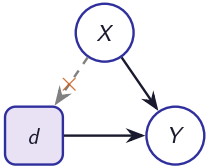
When we *assign* tutoring, preparation no longer decides who gets it: the arrow into D is cut. Pearl writes this intervention $do(D = d)$.

Preparation still affects the score, so the arrow $X \to Y$ stays.

seeing: $D = d$



doing: $do(D = d)$



Remember. A causal effect compares outcomes under $do(D = 1)$ and $do(D = 0)$, not under $D = 1$ and $D = 0$.

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Idea: compare like with like

Building a fair comparison

IDENTIFICATION

If preparation is the only confounder, compare tutored and untutored students with the same preparation.

Preparation group	Share of students	Tutored mean	Untutored mean	Within-group gap
Low	60%	65	55	10
High	40%	85	80	5

Inside each row, preparation cannot explain the gap, because everybody has the same preparation.

Remember. Split into comparable groups first, compare inside each group second.

Put the groups back together: standardisation

Building a fair comparison

IDENTIFICATION

To answer a question about the whole school, weight each group by its share of the school.

$$\text{ATE} = 0.60 \times 10 + 0.40 \times 5 = 8 \text{ points.}$$

A simple average of the two gaps, $(10 + 5)/2 = 7.5$, would describe an imaginary school with equal numbers in each group.

This is **standardisation**: within-group comparisons, then a weighted average using the target population's composition.

Remember. The weights encode the population the answer is about.

Change the weights, change the question

Building a fair comparison IDENTIFICATION

Among students who actually took tutoring, 80% had low preparation.

Keep the same within-group gaps but use the participants' composition:

$$\text{ATT} = 0.80 \times 10 + 0.20 \times 5 = 9 \text{ points.}$$

Weakly prepared students both benefit more and sign up more. That is why ATT exceeds ATE here. We will prove the exact relationship at the end of the session.

Remember. Same data, same group effects, different target, different number.

The adjustment formula is the table, generalised

Building a fair comparison

IDENTIFICATION

Replace "two preparation groups" by "every possible value of X".

$$E\big[Y(d)\big] = E_X\Big\{\, E\big[Y \mid D = d,\, X\big]\Big\}$$

Inner average: the mean score of students with profile X who received d (one cell of the table). Outer average: combine the cells using the population shares of X (the 60% / 40%).

Then $\text{ATE} = E[Y(1)] - E[Y(0)]$. This is also called the **g-formula** or **g-computation**.

Remember. The left side is causal and unobservable; the right side is computable from data.

Three conditions make the comparison fair

Building a fair comparison IDENTIFICATION

The table only works if three things are true. In plain words first:

1. **No hidden common cause.** Within each preparation group, who got tutoring is unrelated to how students would score. *(exchangeability)*
2. **Someone to compare with.** Each group contains both tutored and untutored students. *(positivity)*
3. **Well-defined treatment.** "Tutored" means the same intervention for everyone, and my tutoring does not change your score. *(consistency, no interference)*

Remember. Next slides: one condition at a time, from words to symbols.

Exchangeability, in symbols

Building a fair comparison

IDENTIFICATION

"Within a group, who got tutoring is unrelated to potential scores" becomes:

$$\big(Y(0), Y(1)\big) \;\perp\; D \;\mid\; X$$

The symbol $\perp$ reads "is independent of". Among students with the same X, tutored and untutored students would have had the same distribution of scores in either world.

Also called **unconfoundedness** or **conditional ignorability**. It cannot be tested from the data: it is justified by knowledge of how treatment was assigned.

Remember. Exchangeability makes the untutored students valid stand-ins for the missing world.

Positivity: someone to compare with

Building a fair comparison

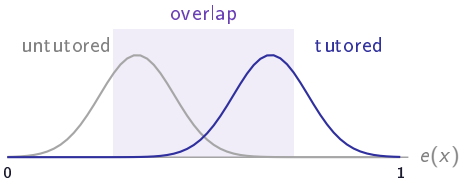
IDENTIFICATION

If every weakly prepared student is tutored, who tells us their score without tutoring?

Let $e(x) = P(D = 1 \mid X = x)$, the **propensity score**. Positivity asks

$$0 < e(x) < 1 \quad \text{for every relevant } x.$$

Very small comparison groups make estimates unstable even before overlap disappears.



Remember. No amount of data can supply a comparison group that does not exist.

Consistency and no interference

Building a fair comparison IDENTIFICATION

The symbols $Y_i(1)$ and $Y_i(0)$ only make sense if "tutoring" is one well-defined thing.

Consistency. If one student had one hour online and another forty hours in person, the label $D = 1$ hides different treatments. We need $Y_i = Y_i(D_i)$ for the treatment we defined.

No interference. If tutored students share their notes with friends, a student's score depends on others' treatment too. Then $Y_i(d_i)$ must be replaced by an outcome indexed by the whole assignment vector.

Together these are often called **SUTVA**.

Remember. Define the treatment carefully before writing any formula.

Proof: where each assumption enters

Building a fair comparison IDENTIFICATION

From intuition: this is exactly the "split, compare, reweight" we did with the table.

Fix $d \in \{0,1\}$. Then

$$\begin{aligned} E[Y(d)] &= E_X\{E[Y(d) \mid X]\} && \text{law of iterated expectations} \\ &= E_X\{E[Y(d) \mid D = d, X]\} && \text{exchangeability} \\ &= E_X\{E[Y \mid D = d, X]\} && \text{consistency.} \end{aligned}$$

Positivity guarantees that $E[Y \mid D = d, X = x]$ is defined for every x in the support of X.

Remember. A proof tells us exactly which assumption carries which step.

What adjustment cannot fix

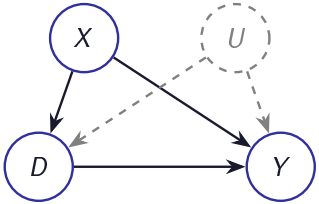
Building a fair comparison

IDENTIFICATION

Motivation U drives tutoring and scores, but the school never measured it.

Two students with the same preparation X can still differ in motivation. The back route $D \leftarrow U \rightarrow Y$ stays open whatever we do with X.

Regression, matching and machine learning on X all inherit this problem: a better fit is not a smaller bias.



Remember. Adjustment removes only the confounding carried by what you measured.

Session 1 roadmap

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7. When assumptions fail	Master 2
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Two stories, identical data

The Master 2 toolkit MASTER 2

From intuition: Awa's missing world, pushed to the extreme.

Every tutored student passes and every untutored student fails.

Story A. Everyone would fail without tutoring and pass with it: the effect is 1 for everybody.
Story B. Strong students choose tutoring and would pass anyway; weak students never come: the effect is 0.

Both stories generate exactly the same spreadsheet. Even infinite data cannot tell them apart.

Remember. More data improves estimation; only assumptions deliver identification.

Identification, formally

The Master 2 toolkit MASTER 2

From intuition: "the data cannot tell the stories apart" becomes a statement about maps between laws.

Let P be a law of the full data $(X, D, Y(0), Y(1))$, $T(P)$ the induced law of the observed (X, D, Y), and $\theta(P) = E_P[Y(1) - Y(0)]$. Within a model $\mathcal{M}$, θ is **identified** if

$$T(P) = T(Q) \implies \theta(P) = \theta(Q) \qquad \text{for all } P, Q \in \mathcal{M}.$$

Equivalently, there is a functional ψ with $\theta(P) = \psi(T(P))$ on $\mathcal{M}$. Under exchangeability, positivity and consistency, ψ is the g-formula.

Remember. Identification is a property of the model and the observables, prior to any estimator.

A formal non-identification construction

The Master 2 toolkit MASTER 2

From intuition: Stories A and B, written as two probability laws.

Let $U \sim \text{Bernoulli}(1/2)$ and $D = U$.

Model A: $Y(0) = 0$, $Y(1) = 1$, so $\text{ATE} = 1$.
Model B: $Y(0) = Y(1) = U$, so $\text{ATE} = 0$.

Under consistency both imply $Y = D$ and $P(D = 1) = 1/2$: the observed laws coincide, $T(P_A) = T(P_B)$, while $\theta(P_A) \neq \theta(P_B)$.

Remember. Without further restrictions, no statistic can distinguish an effect of 0 from an effect of 1.

Regression on a treatment dummy reproduces the raw gap

From intuition: the 75 versus 65 comparison is exactly what OLS computes.

Let $p = P(D = 1) \in (0, 1)$ and $\mu_d = E[Y \mid D = d]$. Then

$$E[DY] = p\mu_1, \qquad E[Y] = p\mu_1 + (1 - p)\mu_0,$$

$$\mathrm{Cov}(D, Y) = p(1 - p)(\mu_1 - \mu_0), \qquad \mathrm{Var}(D) = p(1 - p).$$

Hence the population slope of Y on $(1, D)$ is $\beta = \mu_1 - \mu_0$, whether or not the comparison is causal.

Remember. A regression coefficient is a contrast; causality comes from assumptions, not from OLS.

Omitted-variable bias

The Master 2 toolkit MASTER 2

From intuition: the "head start" due to motivation, as a closed-form bias.

Assume $Y = \theta D + \gamma U + \varepsilon$ with finite second moments, $\mathrm{Var}(D) > 0$ and $\mathrm{Cov}(D, \varepsilon) = 0$. Regressing Y on $(1, D)$ only:

$$\beta_{OLS} = \frac{\mathrm{Cov}(D, Y)}{\mathrm{Var}(D)} = \theta + \gamma \frac{\mathrm{Cov}(D, U)}{\mathrm{Var}(D)}.$$

Proof: substitute the structural equation and use bilinearity of the covariance. The bias has the sign of $\gamma \cdot \mathrm{Cov}(D, U)$: motivated students ($\gamma > 0$) who volunteer ($\mathrm{Cov} > 0$) inflate the estimate.

Remember. Confounding bias is a population quantity, not sampling noise.

Omitted-variable bias after adjusting for controls

The Master 2 toolkit MASTER 2

From intuition: "adjusting for X does not remove motivation", quantified.

Let M_X be population linear residualisation on controls including an intercept, $\tilde{D} = M_X D$, $\tilde{U} = M_X U$. In

$$Y = \theta D + X'\beta + \gamma U + \varepsilon, \qquad E[\tilde{D}\varepsilon] = 0,$$

the Frisch–Waugh–Lovell theorem gives

$$\beta_{D|X} = \theta + \gamma \frac{E[\tilde{D}\tilde{U}]}{E[\tilde{D}^2]}.$$

Only the part of U not linearly explained by X still biases the estimate.

Remember. Good controls shrink the bias; they only remove it if $\tilde{U}$ is uncorrelated with $\tilde{D}$.

ATE, ATT and ATU as weighted averages

The Master 2 toolkit MASTER 2

From intuition: $5.5 = \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 6$ in the oracle table, and the 60/40 versus 80/20 weights.

By iterated expectations over D,

$$\text{ATE} = p\,\text{ATT} + (1-p)\,\text{ATU}.$$

By iterated expectations over X, and under conditional exchangeability,

$$\text{ATE} = \int \tau(x)\, dF_X(x), \qquad \text{ATT} = \int \tau(x)\, dF_{X|D=1}(x).$$

Same function τ, different integrating measures. They differ whenever assignment depends on X and τ varies with X.

Remember. Target populations are integrating measures.

Mean exchangeability is enough for a mean effect

The Master 2 toolkit MASTER 2

From intuition: we only used averages in the table, never full distributions.

Full independence $(Y(0), Y(1)) \perp D \mid X$ is sufficient but stronger than needed for ATE. It suffices that for each d

$$E[Y(d) \mid D = d, X] = E[Y(d) \mid X].$$

With consistency and positivity this identifies $E[Y(d)]$. Identifying the whole distribution $F_{Y(d)}$ requires the analogous distributional restriction.

Remember. Match the strength of the identifying assumption to the target functional.

G-computation for a whole distribution

The Master 2 toolkit MASTER 2

From intuition: apply the same table trick to "score below y" instead of the score.

Under exchangeability, consistency and positivity, take $h_y(Y) = 1\{Y \leq y\}$:

$$F_{Y(d)}(y) = E_X\{P(Y \leq y \mid D = d, X)\}.$$

Marginal quantiles follow as $Q_d(u) = \inf\{y : F_{Y(d)}(y) \geq u\}$.

Caution: $Q_1(u) - Q_0(u)$ compares two marginal distributions. It does *not* identify the distribution of individual effects $Y(1) - Y(0)$, which depends on the unobserved joint law.

Remember. One identification argument serves many functionals of potential outcomes.

ATT by standardisation

The Master 2 toolkit MASTER 2

From intuition: for ATT we only need the tutored students' missing world, 70 in the oracle table.

Write $\text{ATT} = E[Y \mid D = 1] - E[Y(0) \mid D = 1]$. Only the second term is counterfactual. Under untreated-outcome mean exchangeability,

$$E[Y(0) \mid D = 1] = E\{m_0(X) \mid D = 1\}, \qquad m_0(x) = E[Y \mid D = 0, X = x].$$

This requires untutored students on the *tutored* group's covariate support: $e(x) < 1$ there, but $e(x) > 0$ everywhere is not needed.

Remember. ATT needs weaker overlap and weaker exchangeability than ATE.

ATT by inverse-odds weighting

The Master 2 toolkit MASTER 2

From intuition: instead of predicting, reweight the untutored students to look like the tutored ones.

Let $p = P(D = 1)$ and $e(X) = P(D = 1 \mid X)$. Conditioning on X,

$$E\left[(1 - D)\frac{e(X)}{1 - e(X)}\, Y\right] = E\big[e(X)\, m_0(X)\big] = p\, E\big[m_0(X) \mid D = 1\big],$$

where the last step uses $dF_{X|D=1} = \frac{e(x)}{p}\, dF_X$ (Bayes). Therefore

$$\text{ATT} = \frac{1}{p}\, E\left[D\, Y - (1 - D)\frac{e(X)}{1 - e(X)}\, Y\right].$$

Remember. Odds weights turn the control distribution of X into the treated one.

Weak overlap: identified but badly estimated

The Master 2 toolkit MASTER 2

From intuition: "very small comparison groups make estimates unstable", made precise.

Strict positivity $0 < e(X) < 1$ a.s. gives point identification. Estimation needs more. Under **strong overlap** $\epsilon \leq e(X) \leq 1 - \epsilon$ and $E[Y^2] < \infty$,

$$E\left[\frac{D\,Y^2}{e(X)^2}\right] \leq \frac{E[Y^2]}{\epsilon^2},$$

so inverse-probability weights have finite variance. Without a uniform bound, identification can hold while the variance of the weighting estimator is infinite and $\sqrt{n}$-rates fail.

Remember. Identified does not mean estimable at the usual rate.

Uncertainty for plug-in standardisation

The Master 2 toolkit MASTER 2

From intuition: the table's cell means are themselves estimated.

Suppose $m_d(x;\beta)$ is correctly specified and a fixed target law Q defines

$$\theta(\beta) = \int \left[m_1(x;\beta) - m_0(x;\beta)\right] dQ(x).$$

If $\sqrt{n}(\hat{\beta} - \beta_0) \Rightarrow N(0, \Sigma)$, the delta method yields

$$\sqrt{n}\{\theta(\hat{\beta}) - \theta(\beta_0)\} \Rightarrow N(0, a'\Sigma a), \qquad a = \nabla_\beta \theta(\beta_0).$$

If Q is the empirical distribution of X in the same sample, its sampling variability adds a second term.

Remember. Standardisation propagates both model and target-population uncertainty.

Causal graphs as probability: Markov factorisation

The Master 2 toolkit MASTER 2

From intuition: the arrows we drew become a product of conditional densities.

In an acyclic structural model with variables in topological order,

$$V_j = f_j(V_{\mathrm{pa}(j)}, U_j), \qquad U_1, \dots, U_K \text{ mutually independent.}$$

Each V_j is independent of its non-descendants given its parents, so the chain rule collapses to

$$p(v_1, \dots, v_K) = \prod_{j=1}^{K} p\big(v_j \mid v_{\mathrm{pa}(j)}\big).$$

Hidden common causes (U_j dependent) break this factorisation over the observed variables.

Remember. The graph encodes conditional independences the data must satisfy.

Intervention as truncated factorisation

The Master 2 toolkit MASTER 2

From intuition: "cut the arrow into D" from the seeing-versus-doing slide.

In a causally sufficient acyclic model, $do(D = d)$ replaces the factor $p(d \mid v_{\mathrm{pa}(D)})$ by a point mass at d:

$$p\big(v_{-D} \mid do(d)\big) = \prod_{j:\, V_j \neq D} p\big(v_j \mid v_{\mathrm{pa}(j)}\big)\Big|_{D=d}.$$

With graph $X \to D \to Y$, $X \to Y$, integrating out X gives

$$p\big(y \mid do(d)\big) = \sum_x p(y \mid d, x)\, p(x), \qquad \text{not} \qquad p(y \mid d) = \sum_x p(y \mid d, x)\, p(x \mid d).$$

Remember. The g-formula and the do-calculus reach the same adjustment from two directions.

Collider bias in closed form

The Master 2 toolkit MASTER 2

From intuition: the violet dots in the admission scatter plot.

Let X, U be independent $N(0,1)$ and $S = X + U$. For jointly Gaussian $A = (X, U)'$ and S, the Schur complement gives

$$\mathrm{Var}(A \mid S) = \Sigma_{AA} - \Sigma_{AS}\Sigma_{SS}^{-1}\Sigma_{SA} = I_2 - \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

So $\mathrm{Cov}(X, U \mid S) = -\frac{1}{2}$ although $\mathrm{Cov}(X, U) = 0$.

Remember. Conditioning on a common effect subtracts the variation the total explains, creating dependence.

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Sensitivity analysis: how wrong could we be?

When assumptions fail

MASTER 2

From intuition: "motivation was never measured" — so how much would it matter?

Instead of only saying "there may be hidden confounding", quantify it. Suppose the adjusted gain is 8 points. Ask: how large would the remaining motivation gap between tutored and untutored students have to be to bring the effect to zero?

With the OVB formula, bias $= \gamma \cdot E[\tilde{D}\,\tilde{U}]/E[\tilde{D}^2]$: choose plausible values of both factors and recompute.

Remember. Report which violations would change the decision, not just that violations are possible.

Worst-case bounds with no assumption on assignment

When assumptions fail MASTER 2

From intuition: fill each "?" in the school's table with the most extreme possible value.

Assume only consistency and $0 \leq Y(d) \leq 1$ (for example, pass/fail). Let $p = P(D = 1)$ and $\mu_d = E[Y \mid D = d]$.

$$E[Y(1)] = p\mu_1 + (1-p)\underbrace{E[Y(1) \mid D = 0]}_{\in [0,1]} \in \big[p\mu_1,\ p\mu_1 + 1 - p\big],$$

$$E[Y(0)] \in \big[(1-p)\mu_0,\ (1-p)\mu_0 + p\big].$$

No assumption on how treatment was assigned has been used.

Remember. Bounded outcomes give an identified interval even when a point is out of reach.

Sharp bounds on the ATE

When assumptions fail MASTER 2

From intuition: the widest and narrowest answers compatible with the half-empty table.

Subtract the largest possible $E[Y(0)]$ from the smallest $E[Y(1)]$, and conversely:

$$L = p\mu_1 - (1-p)\mu_0 - p, \qquad U = p\mu_1 + 1 - p - (1-p)\mu_0.$$

Set every missing $Y(1)$ to 0 and every missing $Y(0)$ to 1 to attain L; reverse to attain U. Both completions reproduce the observed data, so the bounds are **sharp**.

Example: $(p, \mu_1, \mu_0) = (0.4,\ 0.7,\ 0.5)$ gives $[L, U] = [-0.42,\ 0.58]$. Width is always 1.

Remember. Without assumptions the data cannot even sign the effect.

Monotone treatment response tightens the bounds

When assumptions fail MASTER 2

From intuition: "tutoring never hurts" is an assumption with a price and a payoff.

Assume $Y_i(1) \geq Y_i(0)$ for every unit, with $Y(d) \in [0,1]$. Then $\text{ATE} \geq 0$, and 0 is attained by filling each missing outcome with the observed one ($Y_i(1) = Y_i(0)$). The completion giving U also satisfies monotonicity. Hence

$$\text{ATE} \in [0,\ U].$$

This restriction concerns outcomes; it is different from the monotone treatment *choice* used later for instrumental variables.

Remember. Every tighter bound must be paid for with an explicitly stated restriction.

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Discussion

Practice and wrap-up

SYNTHESIS

Two programmes show the same 10-point gap. One assigned tutoring by lottery, the other took volunteers.

1. Why can the same number carry different causal credibility?
2. In a tutoring study of your own design, name one confounder, one mediator and one collider.
3. Which of the three would you adjust for if you want the total effect?

Remember. Numerical equality does not imply equal evidence.

Exercise: whose effect, and which controls?

Practice and wrap-up

SYNTHESIS

Pairs, ten minutes. Use the preparation table (low: 65 vs 55, share 60%; high: 85 vs 80, share 40%).

1. Compute the ATE.
2. If 80% of participants have low preparation, compute the ATT with the same group effects.
3. Should you adjust for study habits measured *after* tutoring? Why?
4. Which key assumption can never be checked in the data?

Remember. Explain in words before doing the arithmetic.

Exercise: solution

Practice and wrap-up

SYNTHESIS

1. $\text{ATE} = 0.6 \times 10 + 0.4 \times 5 = 8$.
2. $\text{ATT} = 0.8 \times 10 + 0.2 \times 5 = 9$.
3. No: study habits are a mediator. Adjusting for them removes part of the total effect.
4. Exchangeability within baseline groups. Positivity can be inspected; consistency is a design question.

Remember. The target population and the control set determine what the estimate means.

Proof exercise: why ATT exceeded ATE

Practice and wrap-up SYNTHESIS

From intuition: 9 versus 8 in the exercise: weakly prepared students gained more *and* signed up more.

Let $\tau(X) = E[Y(1) - Y(0) \mid X]$, $e(X) = P(D = 1 \mid X)$, $p = E[e(X)]$, and assume conditional exchangeability. Prove

$$\text{ATT} - \text{ATE} = \frac{\text{Cov}\left(\tau(X),\, e(X)\right)}{p}.$$

Then (i) state when ATT and ATE coincide, and (ii) explain why zero covariance does not imply constant individual effects.

Remember. Selection towards profiles with larger effects raises the participants' average.

Proof exercise: solution

Practice and wrap-up SYNTHESIS

By Bayes' rule $dF_{X|D=1} = \frac{e(x)}{p} dF_X$, so

$$\text{ATT} = \frac{E[e(X)\,\tau(X)]}{p}.$$

Since $\text{ATE} = E[\tau(X)]$ and $p = E[e(X)]$,

$$\text{ATT} - \text{ATE} = \frac{E[e\tau] - E[e]\,E[\tau]}{p} = \frac{\text{Cov}(e, \tau)}{p}.$$

Check on the table: $e(\text{low}) = \frac{0.8p}{0.6}$, $e(\text{high}) = \frac{0.2p}{0.4}$ gives $\text{Cov}(e, \tau)/p = 1 = 9 - 8$.
(i) Equality iff the covariance is zero. (ii) Effects may vary within profiles, or across profiles orthogonally to e.

Remember. ATE = ATT under effect–selection orthogonality, not only under constant effects.

What we can now say, step by step

Practice and wrap-up

SYNTHESIS

Intuition	A raw gap mixes a treatment effect with a head start; the missing world is never seen.
Notation	$Y(1), Y(0)$; the switch $Y = DY(1) + (1-D)Y(0)$; ATE, ATT, CATE; gap = ATT + selection bias; confounder, mediator, collider.
Identification	Exchangeability, positivity and consistency turn the g-formula into a causal answer.
Master 2	Identification as invariance; OVB; inverse odds; overlap and variance; truncated factorisation; sharp and monotone bounds.

Remember. A causal method earns its interpretation through its design and its assumptions.

Next session: can a lottery create a fair comparison?

Practice and wrap-up SYNTHESIS

Today the selection-bias term was the enemy. Next week we remove it by design.

If tutoring places are allocated by lottery, assignment cannot depend on how students would score:

$$D \perp \big(Y(0), Y(1)\big) \implies E[Y(0) \mid D = 1] = E[Y(0) \mid D = 0].$$

We will study what randomisation buys, what chance still changes, and how to design a useful experiment.

Remember. Randomisation makes exchangeability true by construction instead of by argument.

Further reading

Practice and wrap-up SYNTHESIS

- Hernán, M. A. and Robins, J. M. (2020). *Causal Inference: What If*. Chapman & Hall/CRC. Ch. 1–4.
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- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer.
- Holland, P. W. (1986). Statistics and causal inference. *JASA* 81(396), 945–960.

Remember. Hernán and Robins, chapters 1–3, is the best companion to today's session.